

Hotel Review Classification Using Tagged Learning and Model Refinement

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Abstract—The increasing use of Natural Language Processing in customer service highlights the importance of understanding how machines process and classify human language. This paper presents a hotel review classification model developed using the Machine Learning for Kids platform. The model was trained to identify three categories: Positive_Praise, Negative_Complaint, and Urgent_Support. Several experiments were conducted using clear review phrases, sarcastic statements, unfamiliar expressions. The initial results showed that the model performed well on direct and familiar phrases but had difficulty with sarcasm, indirect meaning, and out-of-vocabulary inputs. To improve the model, additional and targeted training phrases were added through three refinement iterations. The average confidence score increased from 71.10% in the first iteration to 85.00% in the second iteration and 95.70% in the third iteration. The findings showed that the model became more reliable when the training data were relevant, varied, and correctly tagged. However, the results also showed that correct predictions may still have low confidence when the phrases contain words that can belong to more than one category. Overall, the activity demonstrated that tagged learning and model refinement can improve hotel review classification, but continuous testing and human review are still needed to ensure accurate and meaningful results.

Keywords—Natural Language Processing, sentiment analysis, hotel reviews, tagged learning, text classification, model refinement

I. INTRODUCTION

Natural Language Processing (NLP) has different branches, and one of these is sentiment analysis. Sentiment analysis is a branch of NLP that enables computers to examine and categorize text based on the emotions, opinions, and intentions expressed in written statements. In the hotel industry, it can be used to analyze customer reviews about their experiences, such as the quality of the rooms, staff service, cleanliness, facilities, food, and overall satisfaction.

This experiment uses the Machine Learning for Kids platform, which allows users to train and test a machine-learning model using labeled hotel review phrases. The model is designed to classify customer reviews into three categories: Positive_Praise, Negative_Complaint, and Urgent_Support. Positive_Praise includes reviews that express satisfaction or appreciation, Negative_Complaint refers to reviews that show disappointment or dissatisfaction, and Urgent_Support covers concerns that

require immediate attention, such as safety, health, payment, or security issues. Through this experiment, the model's ability to recognize the sentiment and intent of customers based on their hotel experiences can be tested and evaluated.

II. METHODOLOGY

A. Tool Setup & Interface Overview

To begin with the experiment, the Machine Learning for Kids platform was used to provide hands-on experience in understanding how sentiment analysis works. The platform also offers free access to users who want to explore and develop basic machine-learning models without creating an account.

The platform presented three main buttons: Train, Learn & Test, and Make. The Train section allowed users to input and label the initial hotel review phrases according to their appropriate categories. The Learn & Test section enabled to train the selected model and evaluate it using new test phrases. Lastly, the Make section allowed the trained model to be transferred and integrated into another application or used to create an interactive project, such as a game.

B. Data Collection & Tagging

The dataset was prepared using hotel customer feedback that represented different types of experiences. The collected phrases were organized into three tagging: Positive_Praise, Negative_Complaint, and Urgent_Support. Ten (10) distinct review phrases were placed under each tag, resulting in thirty (30) labeled examples for the initial training dataset. Reviews expressing satisfaction with the hotel's services, rooms, staff, or facilities were tagged as Positive_Praise. Statements describing poor service, inconvenience, or dissatisfaction were labeled as Negative_Complaint, while concerns involving safety, health, security, payments, or other situations requiring immediate attention were tagged as Urgent_Support. After the phrases were reviewed and assigned to their appropriate categories, they were entered into the Train section of the Machine Learning for Kids platform.

C. Model Training

After the 30 labeled hotel review phrases were entered into the Train section, the Learn & Test stage of the Machine Learning for Kids platform was used to train the model. The platform analyzed the ten (10) examples under each category—Positive_Praise, Negative_Complaint, and Urgent_Support—to identify the words and language patterns associated with each class. Once the training

process was completed, 10 new hotel review phrases that were not included in the original dataset were entered individually into the testing area. For each phrase, the predicted category and confidence score were recorded and compared with the expected category. Incorrect predictions were examined to determine whether the model was confused by overlapping meanings, sarcasm, or unfamiliar vocabulary. These observations were then used as the basis for adding new training examples and improving the model's classification performance.

D. Testing & Stress Testing the Model

Based on the experiment summary presented in Table I, a total of 30 test phrases were generated to evaluate the classification performance and confidence of the hotel review sentiment model. The test phrases were different from the examples previously provided to the model during training, allowing the experiments to measure how well the model handled new and unseen inputs.

TABLE I. EXPERIMENT SUMMARY

Experiment Summary		
Experiment	Total Test Cases	Average Confidence Score
1st Experiment (Expected Success)	10	77.50%
2nd Experiment (Sarcastic)	10	57.40%
3rd Experiment (Out-of-the-Vocabulary)	10	31.90%

The first experiment, which used clear and direct hotel reviews, achieved the highest average confidence score of 77.50% and correctly classified all 10 test cases. The second experiment used sarcastic reviews and obtained an average confidence score of 57.40%, with only 5 out of 10 phrases classified correctly.

This showed that the model was often misled by positive words used in a negative or urgent context. The third experiment used neutral and out-of-vocabulary hotel-related statements and recorded the lowest average confidence score of 31.90%. Since these phrases did not clearly belong to any of the three available categories, the model still assigned them a label despite its uncertainty. Overall, the results showed that the model performed well on clear and familiar patterns but became less reliable when processing sarcasm, indirect wording, and unfamiliar expressions. This reflects the importance of using more training data to improve performance.

E. Model Refinement Iteration

The model refinement process was conducted to improve the model's ability to distinguish among Positive_Praise, Negative_Complaint, and Urgent_Support, when the test phrases contained sarcasm, unfamiliar expressions, indirect meanings, or emergency-related words. Each refinement iteration was based on the weaknesses

observed in the previous testing phase. Instead of simply increasing the amount of training data, new phrases were selected according to the specific patterns that caused incorrect predictions or low confidence scores.

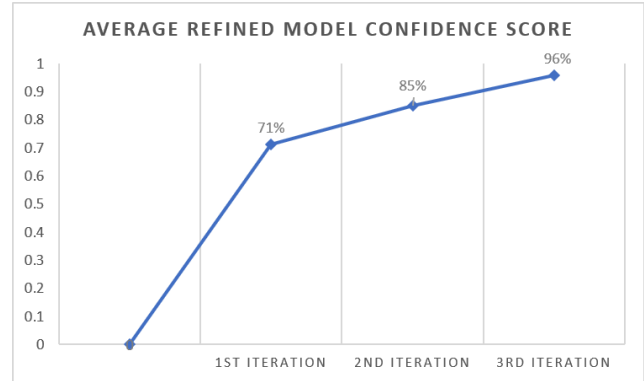


Fig. 1. Average Refined Model Confidence Score

In the first iteration, five new training phrases were added to each category. These examples focused on sarcasm and indirect wording so the model would consider the complete meaning of a phrase instead of relying only on positive or negative words. This iteration produced an average confidence score of 71.10%. Although many predictions were correct, the model still struggled with sarcastic complaints and phrases containing emergency-related words.

In the second iteration, ten additional phrases were added to each category. The new data included clearer examples of positive hotel experiences, ordinary complaints, and emergency situations. The result had increased to 85.00%, and all ten test phrases were classified correctly. However, several Negative_Complaint cases still had lower confidence, particularly those involving a warning tone, a shaking elevator, and a loose door handle.

The third iteration focused only on Negative_Complaint because some phrases were still being confused with Urgent_Support. Ten new phrases were added and the result increased to 95.70% and correctly classified all ten new Negative_Complaint test phrases. The results show that adding targeted training data based on previous weaknesses improved the model's confidence and its ability to distinguish complaints from urgent situations.

III. DISCUSSIONS

A. Explicit Rules vs. Tagged Learning

This kind of activity is not limited to machine learning. The use of an IF-ELSE statement is possible, but it is not advisable because each customer has different forms of complaints. Some use jargon words, sarcasm, unpredictable words, and even expressions. From the earlier experimentation, we used some confusing phrases. A phrase may start with praise, but its whole thought is negative. Because of this, an IF-ELSE statement is not suitable for this kind of behavior because it may fail to understand the real meaning, and it would also be harder to code and manage all the possible exceptions needed to predict accurately. To solve the complexity of customer complaints, tagged learning is more suitable because the model can learn patterns from the training data instead of depending on the rule-based conditions of IF-ELSE.

B. Data Bias & Dataset Quality

Data quality and quantity are the most important things when it comes to deep learning. Earlier in the activity, 10 initial data were trained and tested. Despite having high accuracy on the expected success cases, there was a huge difference when the model was tested using sarcastic phrases. This behavior shows that the trained dataset lacked enough quantity and variety. Moreover, the quality of the data also matters because the phrases must be correctly tagged, related to the categories, and different from one another. If the data are repeated, unclear, or incorrectly labeled, adding more of them may not improve the model. After this observation, we added 15 more phrases and conducted two more refinement iterations. The results greatly increased because the model received more examples to learn from.

Aside from quantity, the quality of the data also matters because data must be related to the categories. If the data are incorrectly labeled then the model may give a wrong prediction. Therefore, the model does not become better only because it has more data, but because it has relevant, varied, and correctly tagged training data.

C. Connecting to the Slide Architecture

Based on the activity, the feature extraction most likely happened after the customer review was entered for training but before the model made its prediction. Before the model generates its prediction, the model first identifies the patterns that were analyzed during the training for the tagged learning and identifies important words on the data. These features are then converted to a computer language (numerical) to understand human language. After the computer understands and analyzes. It will predict whether the review belongs to your categories which are Positive_Praise, Negative_Complaint, or Urgent_Support based on the learned pattern during the training.

IV. CONCLUSION

Through this activity, I learned how Natural Language Processing and tagged learning can be used to understand and classify hotel customer reviews. As I observed, the model performed well on clear and direct phrases with an average of confidence score of 71%, but it became confused when the reviews used sarcasm, indirect meaning, and unfamiliar expressions resulting in the confidence score to decline to 57% and 31%. This result illustrated that a model may give correct results in simple cases but may struggle when the language becomes more complex as the model was not trained enough to encounter this sarcastic and indirect forms of reviews.

The refinement process also helped me understand the importance of checking incorrect predictions and low confidence scores. By adding relevant, varied, and correctly tagged training phrases, the model's average confidence improved from 71.10% to 95.70%. From this experience, I realized that improving a model is not only about adding more data, but also about adding the right kind of data based on the weaknesses found during testing. Overall, this activity gave me a clearer understanding of how training data, testing, and continuous refinement work together to create a more reliable text classification model.